Minor Project Synopsis Report

ATTENDANCE SHORTAGE CALCULATOR

Project Category: University Based Project

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To

K.R Mangalam University

by

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ABSTRACT

- Attendance is critical for academic performance and exam eligibility.
- Many students fail to meet minimum attendance due to lack of real-time tracking.
- Attendance records are often manual or fragmented, making monitoring difficult.
- The project proposes a **web-based Attendance Shortage Calculator**.
- Students can enter subject-wise attendance details.
- The system calculates attendance percentage instantly.
- It predicts future attendance shortages in advance.
- It also calculates extra lectures needed to recover shortages.
- The system reduces academic risk and promotes responsible attendance.
- It provides a simple, user-friendly platform for attendance planning.

Keywords: Attendance Management, Web Application, Academic Planning, Flask, Student Analytics

INTRODUCTION

- Attendance monitoring is a mandatory academic requirement.
- Most universities require a minimum of **75% attendance**.
- Low attendance can lead to detention, penalties, or exam disqualification.
- Many institutions use traditional or semi-digital systems.
- These systems lack real-time analysis and predictive insights.
- Students often realize shortages only during exam time.
- Late awareness gives no scope for corrective action.
- There is a growing demand for smart academic monitoring tools.
- The Attendance Shortage Calculator helps track and predict attendance early.

MOTIVATION

- Attendance shortage is a common issue among students.
- The issue often arises due to lack of awareness, not negligence.
- Students are informed about shortages very late.
- Late information increases stress and academic risk.
- The project aims to provide timely attendance updates.
- Real-time calculation helps students plan better.
- Predictive insights allow early corrective action.
- The system promotes accountability and discipline.
- It reduces dependency on manual calculations.
- It ensures transparency and student-centric tracking.

LITERATURE REVIEW

- Various attendance systems exist like RFID and biometric systems.
- Mobile-based attendance systems are also widely used.
- Most systems focus only on attendance collection.
- Data interpretation and prediction are often missing.
- Predictive analysis for attendance shortage is rarely provided.
- Many systems are institution-centric, not student-centric.
- Studies emphasize the importance of early academic monitoring.
- Predictive tools help reduce academic failure.
- Limited work exists on attendance recovery calculation.
- This project focuses on predicting shortages and recovery methods.

GAP ANALYSIS

- Existing systems only display raw attendance data.
- Real-time attendance percentage calculation is missing.
- Shortage prediction is not available.
- No guidance for attendance recovery is provided.
- Student self-monitoring dashboards are limited.
- The proposed system fills these gaps.
- It provides predictive, interactive, and student-friendly features.

PROBLEM STATEMENT

- Students miss minimum attendance due to delayed awareness.
- Current systems do not predict attendance shortages.
- Recovery guidance is not provided.
- Students face academic penalties and exam disqualification.
- A system is needed to calculate, predict, and guide attendance recovery.

OBJECTIVES

- To calculate current attendance percentage accurately.
- To verify eligibility based on minimum attendance criteria.
- To predict attendance shortages in advance.
- To calculate required classes to recover attendance.
- To reduce academic risk through early planning.

TOOLS / TECHNOLOGIES USED

Front-End

- HTML5
- CSS3
- JavaScript

Back-End

- Python
- Flask Framework

Database

- MySQL

METHODOLOGY

- The system follows a modular and user-centric approach.
- Users input subject-wise lecture details.
- Total lectures and attended lectures are entered.
- Attendance percentage is calculated automatically.
- Results are compared with minimum requirements.
- Shortage is identified if criteria are not met.
- The system predicts the number of lectures needed for recovery.
- Results are displayed on a simple dashboard.
- Flask handles backend processing.
- Data storage is managed using databases.